satisfied, even when the extra dimensions are tightly curled up. Second, just as the vibrational patterns of air streams blown through a tuba are affected by the twists and turns of the instrument, the vibrational patterns of strings are influenced by the twists and turns in the geometry of the extra six dimensions. If you were to change the shape of a tuba by making a passageway narrower or by making a chamber longer, the air's vibrational patterns and hence the sound of the instrument would change. Similarly, if the shape and size of the extra dimensions were modified, the precise properties of each possible vibrational pattern of a string would also be significantly affected. And since a string's vibrational pattern determines its mass and charge, this means that the extra dimensions play a pivotal role in determining particle properties.

This is a key realization. *The precise size and shape of the extra dimensionshas a profound impact on string vibrational patterns and hence on particle properties.* As the basic structure of the universe—from the formation of galaxies and stars to the existence of life as we know it—depends sensitively on the particle properties, the code of the cosmos may well be written in the geometry of a Calabi-Yau shape.

We saw one example of a Calabi-Yau shape in Figure 12.9, but there are at least hundreds of thousands of other possibilities. The question, then, is which Calabi-Yau shape, if any, constitutes the extra-dimensional part of the spacetime fabric. This is one of the most important questions string theory faces since only with a definite choice of Calabi-Yau shape are the detailed features of string vibrational patterns determined. To date, the question remains unanswered. The reason is that the current understanding of string theory's equations provides no insight into how to pick one shape from the many; from the point of view of the known equations, each Calabi-Yau shape is as valid as any other. The equations don't even determine the size of the extra dimensions. Since we don't see the extra dimensions, they must be small, but precisely how small remains an open question.

Is this a fatal flaw of string theory? Possibly. But I don't think so. As we will discuss more fully in the next chapter, the exact equations of string theory have eluded theorists for many years and so much work has used *approximate* equations. These have afforded insight into a great many features of string theory, but for certain questions—including the